

Be sure to write all proofs in a neat and logical fashion. Give reasons for all steps!

- ① State and prove the Vitali Convergence Theorem for integrals.
- ② State and prove the Martingale Convergence Theorem.
- ③ Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be finite measure spaces. Construct $\mu \otimes \nu$ on $\mathcal{S} \otimes \mathcal{T}$. Give details.
- ④ Let $\{\mu_n\}$ be a sequence of real valued signed measures on (S, Σ) . Define $\lambda(A) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{|\mu_n|(A)}{1 + |\mu_n|(S)}$, $A \in \Sigma$. Show λ is a finite measure on Σ and $\mu_n \xrightarrow{\Sigma-\mathcal{S}} \lambda$ for each n .
- ⑤ Let $f: [0, b] \rightarrow \mathbb{R}$ be Lebesgue integrable. Suppose $\int_{[0, c]} f \, d\mu = 0$ for each $c \in [0, b]$. Show $f = 0$ a.e. μ .
- ⑥ Let $(\Omega, \mathcal{F}, P)$ be a pr. space, Y an integrable r.v. and $\mathcal{G}$ a sub σ -field of $\mathcal{F}$. Suppose $\mathcal{G}$ and $\mathcal{F}(Y)$ are independent. What is $E(Y | \mathcal{G})$? Prove.
- ⑦ Let $\mathcal{X}$ be a B-space and suppose $\mathcal{Y}$ and $\mathcal{Z}$ are closed linear subspaces of $\mathcal{X}$ such that each $x \in \mathcal{X}$ has a unique representation of the form $x = y + z$, where $y \in \mathcal{Y}$ and $z \in \mathcal{Z}$. Show that there is a constant K such that $\|y\| \leq K\|x\|$ and $\|z\| \leq K\|x\|$, for each $x \in \mathcal{X}$ where y & z are given as above.
 [Hint: define a map $T: \mathcal{X} \rightarrow \mathcal{Y}$ which is useful, use the closed graph theorem to get continuity; then define $S: \mathcal{X} \rightarrow \mathcal{Z}$, etc.]